

Monte Carlo — decision criteria

What this is. A decision procedure. Given the output of `strategies/monteCarlo` for one strategy, it returns exactly one of three verdicts — **ADVANCE**, **KILL**, **HOLD** — together with the rule that decided and the number that fired it. It is written to be executed by a reasoning agent without further judgement calls: every threshold is a stated constant, every constant is quoted with the rate at which it fires on a named reference population, and every formula is closed-form or names the exact key of the result object it reads.

Companion document. `montecarlo-2026-09-13.md` in this folder defines every number the module produces and takes no view on any of them. This document takes the view. Where the two disagree about a formula, that one is the source of truth about what the code computes; this one is the source of truth about what to do with it.

Reference population. Every firing rate, quantile and calibration claim below was measured on the 36 strategies of databank `Results`, project `XAUUSD`, trade export `2026-09-03` (`raw/XAUUSD/Results/2026-09-03/trades/`), whose Monte Carlo run of `2026-09-12` is in `reports/XAUUSD/Results/2026-09-12/montecarlo/`. The per-scope statistics, permutation tests and heterogeneity tests were computed directly from those 36 trade files for this document. One asset, one generation run, one date: the *structure* of the rules is general, the *constants* are calibrated here and must be re-measured when the asset or the generation regime changes. Appendix A is the whole population, strategy by strategy; Appendix B is its quantiles.

Pipeline note. This study runs after the IS/OOS decay test and the cross-market retest. What it may therefore assume, and how every firing rate below changes once that filter is actually applied, is \$11 — read it before using any rate in \$3–\$5 as a expectation.

Scope note. This is not a test of whether an edge exists. It is a test of what an edge rests on, and of whether it survived the walk-forward. Statements about overfitting relative to the number of strategies tried during generation cannot be made here and are not made: that needs the generation population, which does not reach this stage.

0. The contract

Input. One of, in order of preference:

1. `derived/montecarlo/<asset>/<databank>/<strategy>.json` — `body.result` and `body.verdict`, the full result object. This is the only input that carries everything.
2. `reports/.../montecarlo/verdict.csv` plus `flags.csv` — one flat row per strategy. Sufficient for the Layer 0 and Layer 3 reads, **insufficient for Layers 1 and 2**, which need per-scope numbers these files do not carry (§9).
3. The per-strategy HTML report. Human-readable and complete, but the agent must parse it.

Output. For each strategy, one record:

```
decision : ADVANCE | KILL | HOLD
score    : 0-100, meaningful only when no veto fired
binding  : the rule id that decided, e.g. "V2"
value    : the number that fired it
limit    : the constant it was compared against
evidence : every rule that fired, not only the first
risk_scale : the per-trade risk that fits the drawdown budget (§6)
```

Rules the agent may not break.

- **Never report a decision without the rule id and the number.** A verdict whose arithmetic cannot be reproduced from this document is a bug in the agent, not a finding about the strategy.
- **Evaluate every rule, always.** Do not stop at the first veto. Which rules fired *together* is the diagnosis; which fired first is an accident of ordering.
- **A missing number is HOLD, never KILL.** Absence of evidence disqualifies the analysis, not the strategy.
- **Never invent a number that is not in the input.** If a rule needs a statistic the report does not carry, that rule's outcome is HOLD, and §9 names what must be added to the code.

1. The seven principles the rules encode

Read these before the rules. Each rule cites the principle it comes from, and a rule whose principle the agent does not understand will be misapplied on the first strategy that does not resemble the reference population.

P1 — The study measures dependence, not existence

Every family perturbs a backtest that already happened. None of them can create evidence that an edge is real; they can only show what it rests on. So the criteria are built to **disqualify** on demonstrated fragility and to **rank** on everything else. They never certify.

P2 — Only the out-of-sample scope is near-unbiased

These strategies were selected by StrategyQuant X out of a generation population. Selection happened on in-sample behaviour, so `SR_IS` is inflated by an unknown amount and is not an estimate of anything. `SR_OOS` is the near-unbiased estimate. It follows that:

- `SR_OOS < SR_IS` is the **expected** outcome for every selected strategy, not a finding. On the reference population the median retention is 0.136 and 34 of 36 strategies fall below 0.60 — a "warning" that fires on 94% of the population is not a warning (P7).
- The question is never "did it degrade" (it always did). It is **"where did it land, and was the drop larger than sampling noise can explain."** Those are two separate tests, and the rules keep them separate (V1 and V2).
- Any statistic used to *decide* must be computed on the OOS scope, or on both scopes compared correctly. A statistic computed on the pooled stream is dominated by the inflated IS side: with `n_IS / n_OOS ≈ 2.3` on the reference population, a pooled number is roughly 70% in-sample.

P3 — Every cross-scope comparison must be scale-free

On the reference population `n_IS` spans 690–1312 and `n_OOS` spans 292–624. The two scopes are never the same length, so any statistic whose value grows with the sample cannot be compared between them without normalisation.

statistic	how it scales with n	comparable across scopes?
per-trade Sharpe μ/σ	not at all	yes, directly
profit factor	not at all (only its sampling error shrinks)	yes, directly
per-trade expectancy in R	not at all	yes, directly
Ret/DD	weakly (numerator $\propto n$, denominator $\propto \sqrt{n \dots n}$)	only with care
net profit	$\propto n$	no — divide by n
max drawdown <code>dd_pct</code>	grows with n ; $\rightarrow \sigma\sqrt{(\pi n/2)}$ at zero drift	no — see below
longest losing run	$\propto \log n$	no

Drawdown is the trap, because it is the number that looks most like a risk measure and is the least comparable. For an additive equity curve with per-trade mean μ and standard deviation σ , the expected maximum drawdown over n trades is $\sigma\sqrt{(\pi n/2)}$ at $\mu = 0$, and tends to $\sigma^2/(2\mu)$ once drift dominates (Magdon-Ismael et al., 2004). Both limits depend on n , so a shorter scope draws down less for arithmetic reasons alone.

Consequence for the IS/OOS overlay figures (`degrade.py` , §8 of the companion document): as the code stands, the IS histogram is built from `n_IS` trades and the OOS histogram from `n_OOS` trades. The OOS side is therefore handicapped *in its own favour* on `dd_pct` and *against itself* on `net` . Reading the two histograms as a like-for-like comparison is wrong in opposite directions depending on which family is on screen. Two ways out, and the agent must know which is in force:

- Preferred, needs a code change (§9.4):** resample the longer scope at the shorter scope's length, so both distributions describe `min(n_IS, n_OOS)` trades. Then they are directly comparable and no normalisation is needed.
- Available today:** ignore the raw overlay for decisions and use only the scale-free per-scope statistics of §7. Read the overlay as a picture, never as a measurement.

P4 — Never put a near-zero quantity in a denominator

This is the single most likely way an agent will mis-kill a good strategy here, so it gets its own principle. On the reference population the median OOS net profit is a small positive number, and several are near zero or negative. Any ratio of the form `something / net_OOS` therefore explodes, flips sign, or saturates, and its threshold stops meaning what it was calibrated to mean.

Four such ratios were tested for this document and **rejected on measurement**:

rejected statistic	kills on the reference population	why it is meaningless
<code>best_OOS_trade / net_OOS > 0.25</code>	50%	small denominator, not concentration
<code>net_OOS without its best 3 trades ≤ 0</code>	69%	same
<code>trimmed_SR_OOS / untrimmed_SR_OOS < 0.5</code>	67%, values spanning −94.7 to +0.64	denominator near zero
<code>net_5 / net_OOS</code> as a score term	median 0.0, bimodal	same

Their repaired forms are V6 and V7. The general repair: **measure concentration against gross profit, and measure edges in absolute terms — never as a ratio to a near-zero net.**

P5 — Drawdown is a sizing parameter, not a quality gate

This is the owner's standing instruction and it overrides any statistical argument for gating on drawdown. A strategy is not disqualified for drawing down more than a prop-firm limit, because the risk per trade is a free parameter chosen later.

Under additive fixed-USD risk, scaling `risk_per_trade` by λ scales every trade's P&L by λ , hence net profit and dollar drawdown by λ , leaving **Ret/DD exactly invariant**. Ret/DD is therefore the quality number that survives resizing, and `dd_pct` is not. The correct output is not a pass/fail on drawdown but the risk level at which the strategy fits a stated budget (§6).

The existing `dd_99` veto (`dd_pct_99 > 10%`) fires on 21 of 36 strategies and contradicts this principle. **It is removed**, and replaced by the §6 sizing output.

P6 — Separate the distribution's own shape from this backtest's luck

`inflation = p95(dd_pct under reordering) / dd_pct(observed)` is meant to measure how lucky the real ordering was. It does not, because it mixes two independent things. Factor it exactly:

```
inflation = p95 / dd_obs = (p95 / p50) × (p50 / dd_obs)
                ^^^^^^^^^      ^^^^^^^^^^^^^^^^^
                skew_ratio    luck_ratio
```

Measured over all 36 strategies with a 5 000-draw exact permutation (`iid_shuffle`), which preserves the trade multiset):

- `skew_ratio = p95/p50` spans **1.535 to 1.622**, mean 1.578, standard deviation 0.019. It is a constant. It is the right-hand skew of the maximum-drawdown distribution itself, and it carries **no information about any strategy**.
- `luck_ratio = p50/dd_obs` spans 0.739 to 1.662, median 1.137. This is the part that is about the strategy.

So `inflation ≈ 1.578 × luck_ratio`, which means the configured watch threshold of 1.5 fires whenever `luck_ratio > 0.95` — on essentially any strategy whose real drawdown was no worse than the median of its own reorderings. It fires on **36 of 36**. The veto threshold of 3.0 requires `luck_ratio > 1.90`, above the population maximum of 1.662, so it fires on **0 of 36**. One threshold is unreachable, the other unavoidable; neither has ever separated anything.

The statistic that does this job properly is the **permutation rank** (V3), which has a known null distribution and needs no calibration at all.

P7 — A threshold is not a threshold until its firing rate is known

Every constant in this document is quoted with the fraction of the reference population it disqualifies. A rule that fires on everything and a rule that fires on nothing are the same rule: neither decides. The same test applies to score terms — **no term of a composite may be constant across the population**. The current system breaks this twice (§8): sub-score D is 0.0 on all 36 strategies and sub-score E is 100.0 on all 36, so 30% of the composite's weight carries zero information, and the tiers built on top of it are unreachable by construction.

2. Notation and shared primitives

Three formulas recur. Define them once.

2.1 The Sharpe estimate and its standard error

For a set of trades with per-trade P&L `x` :

```
SR = mean(x) / std(x, ddof=1)
g3 = skewness(x)                # scipy.stats.skew
g4 = kurtosis(x, fisher=False)  # raw kurtosis; a normal gives 3
V  = 1 - g3*SR + ((g4 - 1)/4)*SR**2  # Mertens / Lo variance factor
SE = sqrt(V / n)
t  = SR / SE
```

`V` is **already computed by the module**, inside `significance.psr()`, as the `var` term of the Probabilistic Sharpe Ratio. The PSR and the standard error of a Sharpe ratio are the same piece of mathematics: $\text{PSR}(b) = \Phi((SR - b) \cdot \sqrt{(n-1) / V})$. Everything this document does with per-scope Sharpe ratios reuses that one function, applied to a scope instead of to the pooled stream.

`V` corrects for the shape of the P&L distribution. On the reference population `g4` reaches 8.8, so the correction is not cosmetic: fat tails inflate the uncertainty of a Sharpe estimate, and ignoring them makes every test below look more certain than it is.

2.2 Annualisation

Per-trade Sharpe is the scale-free quantity, but it is unreadable. Annualise **per scope**, on that scope's own trade rate:

```
tpy(scope) = n(scope) / years(scope)  # years between that scope's first and last open time
ann_SR(scope) = SR(scope) * sqrt(tpy(scope))
```

Never annualise the OOS scope with the pooled trade rate. On the reference population `tpy` spans 66–129, so using the wrong rate misstates the annualised number by up to 40%. Every threshold below that mentions an annualised Sharpe assumes this definition.

2.3 The score curve

```
curve(v, bad, good) = clip(100 * (v - bad) / (good - bad), 0, 100), and 0 when v is NaN
```

Straight-line, clipped at both ends; `bad` may be above or below `good`. Identical to `scoring.curve()` in the code, including the NaN rule: a missing number scores zero rather than propagating.

3. Layer 0 — admissibility

These checks ask whether the *analysis* is fit to decide on. Every failure here returns **HOLD** and names the remedy. **None of them may ever return KILL** — a strategy is not at fault for a thin export or an unstable simulation.

id	check	formula	threshold	remedy on failure
A1	both scopes present	<code>stream.samples()</code> returns non-empty <code>IS</code> and <code>OOS</code>	both ≥ 1	re-export with the OOS window attached
A2	OOS large enough to decide	<code>n_OOS</code>	≥ 200 to decide; 60–199 $\rightarrow$ decide with <code>HOLD-THIN</code> ; $< 60 \rightarrow$ HOLD	extend the OOS window or retest
A3	OOS percentiles readable	<code>confidence.percentile(n_OOS, 5)</code>	must be <code>reliable</code> , i.e. <code>n_OOS</code> ≥ 200	as A2
A4	block methods have blocks	<code>confidence.blocks(n, min(blocks), 20)</code>	<code>reliable</code>	shorten <code>block_min</code> or accept Family A only
A5	simulation noise is below the decision margin	<code>stability.worst_spread</code>	≤ 0.10	raise <code>global.n_sims</code> and re-run
A6	reordering models are sound	<code>A.invariant[label]</code> for every model in <code>KEEPS_MULTISSET</code>	≈ 0 (float error only)	a bug in the model — stop, do not decide
A7	volatility data covers the trades	<code>D.regime.coverage</code>	≥ 0.95	re-export bars; Family D unusable below this
A8	the asset file matches the backtest	<code>cost_check.diverges</code>	false	Family C and V4 $\rightarrow$ HOLD; the rest still decides

A2 in detail. The threshold is not arbitrary. `confidence.percentile(n, q)` calls a percentile reliable when the expected count in the tail, $n \cdot \min(q, 100-q)/100$, reaches 10. For the 5th percentile that is `n` ≥ 200 . Below 200 trades the OOS 5th percentile rests on fewer than ten observations and V7's and S3's numbers are noise. On the reference population `n_OOS` is 292 at minimum, so A2 never fires there — it exists for the strategies that will arrive with a shorter retest window.

A6 in detail. `iid_shuffle` and `block_shuffle` only permute trades, so net profit is identical in every path by construction and its standard deviation across paths must be zero to floating-point error. A non-zero value is never a property of the strategy; it means the draw model is broken and every number downstream of it is void. This is the one check that stops the analysis rather than downgrading it.

4. Layer 1 — the seven vetoes

Each veto is defensible on its own, fires on a measured fraction of the reference population, and comes with the false-kill risk it carries. A strategy that fires **any** veto is **KILL**.

Ordering is for reading only; the agent evaluates all seven and reports all that fired.

V1 — There is no out-of-sample edge to begin with

Fires when: $SR_{OOS} \leq 0$ **or** $PF_{OOS} \leq 1.0$
Reference population: kills 10 of 36 (27.8%)
Principle: P2

This is a sign test, not a significance test. It asks only that the near-unbiased estimate sit on the correct side of zero. If the out-of-sample Sharpe is negative there is no positive edge for degradation to even be about, and no amount of robustness elsewhere repairs it.

The two conditions are near-equivalent by construction and are both stated so that a scope with an unusual win/loss shape cannot slip through one of them. On the reference population they select the same 10 strategies.

False-kill risk, computed. For a strategy whose *true* annualised OOS Sharpe is S , over n_{OOS} trades at t_{py} trades per year, the estimate has standard error $\approx 1/\sqrt{n_{OOS}}$ in per-trade units, so

$$P(SR_{OOS} \leq 0) = \Phi(-S / \sqrt{t_{py} \cdot \sqrt{n_{OOS}}}) = \Phi(-S \cdot \sqrt{years_{OOS}})$$

With the reference population's 5 OOS years: a truly excellent strategy ($S = 1.0$) is wrongly killed 1.3% of the time; a good one ($S = 0.75$) 4.7%; a mediocre one ($S = 0.30$) 25%. **The rule is deliberately harsh on mediocre strategies and nearly harmless to good ones**, which is the correct asymmetry when the next stage costs real work.

Why not a significance test here. Requiring $t_{OOS} \geq 1.645$ instead would kill 34 of 36 (94.4%). That is not a standards choice, it is a power failure: with $n_{OOS} \approx 440$, reaching $t = 1.645$ demands a true annualised Sharpe near 0.75 *and* good luck. The OOS sample cannot certify, so V1 does not ask it to. Significance information is not discarded — it enters the score through S1 and S2, where being short of certainty costs points instead of a life.

V2 — The decay is larger than noise explains *and* it landed somewhere unusable

Fires when: $z_{decay} > 2.326$ ($p < 0.01$, one-sided) **and** $ann_SR_{OOS} < 0.60$
Reference population: kills 11 of 36 (30.6%)
Principle: P2, P3

This is the rule the owner's priority hangs on, so its construction matters more than any other.

$$\begin{aligned}\Delta &= SR_{IS} - SR_{OOS} \\ SE_{\Delta} &= \sqrt{V_{IS}/n_{IS} + V_{OOS}/n_{OOS}} \\ z_{decay} &= \Delta / SE_{\Delta} \\ p_{decay} &= 1 - \Phi(z_{decay})\end{aligned}$$

The two scopes are disjoint chronological blocks, so their estimates are independent and the variances add. Each V is the §2.1 factor computed on that scope's own trades, so each side carries its own skew and kurtosis rather than a shared assumption.

Both conditions are required, and the reason is not a compromise. A drop is only actionable if it lands the strategy somewhere it cannot be used. A strategy going from annualised Sharpe 3.0 to 1.5 has retention 0.50 and an enormous z_{decay} , and is still excellent. **Vetoing on the size of the drop alone punishes strong strategies for having been strong in-sample.** The landing condition is what makes the rule about usability rather than about arithmetic.

Equally, the landing condition alone is not enough: a strategy can land low simply because it was never strong, which is a level problem V1 and the score already handle. V2 exists for the specific pathology of a strategy that *was* strong in-sample, *is* weak out-of-sample, and whose gap is too large to be luck.

On the reference population: z_{decay} has median 1.99 and spans 0.46–2.99; p_{decay} has median 0.023. So the IS→OOS Sharpe drop is significant at the 5% level for more than half the population and at the 1% level for about a third. This is exactly what P2 predicts from selection, and it is why the threshold is set at $p < 0.01$ rather than $p < 0.05$ — at 5% the rule would fire on 67% of the population and stop discriminating.

Why not the retention ratio that the code uses today. $oos_ratio = \text{median}(SR_{OOS} \text{ bootstrap}) / \text{median}(SR_{IS} \text{ bootstrap})$ has four defects, in increasing order of seriousness:

1. It is a ratio of two noisy estimates, with no tractable sampling distribution. The *difference* has one, and SE_{Δ} above is it.
2. It explodes when SR_{IS} is near zero. (Not binding on the reference population, where SR_{IS} spans 0.095–0.200, but it will bind on a different generation run.)
3. Negative values — 10 of 36 here — are a categorically different failure from "OOS is 30% of IS", and putting both on one axis makes the axis meaningless in the middle.

4. **It ignores sample size entirely.** Retention 0.3 on 600 OOS trades and retention 0.3 on 100 OOS trades are very different pieces of evidence, and the ratio cannot tell them apart.

Retention is still worth reporting, and it is worth points in the score (S2), because it is the number a human reads fastest. It is not worth a veto on its own.

Sensitivity — and it runs the opposite way to intuition. Measured on the reference population, the landing floor is nearly inert and the significance threshold is what decides:

landing floor (z held at 2.326)	V2 kills	z_decay threshold (floor held at 0.60)	V2 kills
0.40	10 / 36	1.645 (p < 0.05)	24 / 36
0.50	11 / 36	2.326 (p < 0.01)	11 / 36
0.60	11 / 36	2.576 (p < 0.005)	4 / 36
0.70	11 / 36		
0.80	11 / 36		

The floor does nothing between 0.50 and 0.80 **on this population**, because `z_decay` and `ann_SR_00S` are themselves correlated at $\rho = -0.66$: the strategies whose decay is significant are already the ones that landed low, so the second condition is almost implied by the first here. The floor is still worth stating, and is set at **0.60**, because that correlation is a property of this databank and not a law — a population containing strategies that were very strong in-sample and merely good out-of-sample would separate the two conditions, and the floor is what stops V2 killing them.

The `z_decay` threshold is the constant to guard. At $p < 0.05$ it fires on 24 of 36 and stops discriminating; at $p < 0.005$ it fires on 4 and stops protecting. $p < 0.01$ is the setting, and it is not a free parameter in the way the floor is.

V3 — The backtest's comfort came from the order the trades arrived in

Fires when: `rank(dd_pct | iid_shuffle) ≤ 0.05` **and** `rank(dd_pct | block_shuffle, largest block) ≤ 0.05`

Reference population: kills 3 of 36 (8.3%) on the first leg

Principle: P1, P6

Reads: `A.runs["iid_shuffle"]["dd_pct"]["rank"]` and `A.runs["block_shuffle/≤max>"]["dd_pct"]["rank"]` — **both already in the result object, no code change needed**

`iid_shuffle` preserves the multiset of trades exactly: every path contains every trade once, only the order differs. So

$$\text{rank} = (1/N) \cdot \sum 1[\text{dd_pct}(\text{simulated path}) \leq \text{dd_pct}(\text{observed})]$$

is an **exact one-sided permutation p-value** for the null "the maximum drawdown does not depend on the order the trades arrived in". Under that null it is Uniform(0,1). No calibration is required and no threshold has to be invented: 0.05 means 0.05.

A low rank says the real sequence drew down less than almost any reshuffling of the very same trades. That comfort is not a property the strategy will reproduce, because the ordering will not repeat.

The second leg separates luck from structure, and it is the reason the rule has two. A rank that stays low under `block_shuffle` — which permutes blocks of consecutive trades and so preserves local sequencing — means the favourable arrangement is *local and real* (losers genuinely cluster where size is small, say, or the strategy genuinely stands down after a loss). A rank that is low under `iid_shuffle` but rises toward 0.5 under `block_shuffle` means the advantage lived in fine-grained ordering that live trading will not reproduce. Only the second case is luck, and requiring both legs kills only that case.

A population-level finding worth recording. Across the 36 strategies the permutation rank is significantly non-uniform: mean 0.330 against the null's 0.500, and a Kolmogorov–Smirnov test against Uniform(0,1) gives $p = 0.0003$. Eight strategies fall in the lowest decile. The real trade sequences drew down materially less than their reshufflings do, population-wide — consistent with the generation stage having selected on drawdown or on Ret/DD, which preferentially keeps strategies whose actual ordering happened to be kind. **This is a fact about the generation filter, not about any one strategy**, and it is the main reason V3 is worth having at all.

V4 — The result does not survive the costs it was actually charged

Fires when: `cost_cushion < 1.5` **or** `spread_cushion < 2.0`

Reference population: `cost_cushion < 1.5` kills 4 of 36 (11.1%); `spread_cushion < 2.0` kills 0 of 36

Principle: P1

Define, on the whole stream:

```
cost_cushion = net_observed / Σ cost_i      # cost_i recovered as gross - net, per trade
spread_cushion = net_observed / Σ spread_i  # spread_i at the asset's sqx_default spread
```

`cost_cushion` is the multiple by which commission and swap would have to rise before the strategy is flat. It is the cleanest single statement of cost robustness available, and it is far more readable than Family C's `keep`, which reports the effect of one particular multiplier draw.

Both cushions are recoverable in closed form from Family C's existing output, so this rule needs no new numbers if $\Sigma \text{ cost}$ and $\Sigma \text{ spread}$ are not stored. From `cost_shock`, with $m \sim U(1,2)$ applied to the *increment* over the recovered cost:

```
E[net'] = net - C · (E[m] - 1) = net - 0.5 · C
keep    = E[net']/net ⇒ C/net = 2 · (1 - keep)
cost_cushion = net/C = 1 / (2 · (1 - keep_cost_shock))
breakeven cost multiple = 1 + cost_cushion
```

From `spread_widen`, with $m \sim U(1,2)$ applied to the **whole** spread cost:

```
E[net'] = net - S · E[m] = net - 1.5 · S
keep    ⇒ S/net = (1 - keep)/1.5
spread_cushion = net/S = 1.5 / (1 - keep_spread_widen)
```

A modelling asymmetry the agent must know about. `cost_shock` subtracts only the *increment* over the cost SQX booked, because the P&L is already net of it. `spread_widen` subtracts the **entire** modelled spread cost, on top of a P&L in which the spread is already inside the fill prices (`core.trades.cost`: "Spread is already inside the fill prices and does not appear here"). So at a multiplier of 1.0 `spread_widen` is **not** the identity — it charges one full extra spread. This is deliberate (a fixed backtest spread is optimistic every single time), but it means `keep_spread_widen` is biased low by roughly S/net and **is not comparable to** `keep_cost_shock`. Compare cushions, which are on a common scale, never the two `keep` s.

On the reference population `cost_cushion` spans 1.08–3.57 with median 2.33, and `spread_cushion` spans 2.70–6.12 with median 4.45. So the median strategy's entire net profit is only 2.3 times the commission and swap it paid — thin, and worth knowing. The `spread_cushion < 2.0` leg fires on nothing here; it is retained as a guard for assets with a wider relative spread than XAUUSD, and its firing rate on this population is stated as zero so that nobody later mistakes it for an active filter (P7).

V5 — The edge is not the same edge in every period

Fires when: Cochran's Q gives $p < 0.05$ and $I^2 > 0.50$

Reference population: kills 5 of 36 (13.9%)

Principle: P1, P7

Needs: per-block Sharpe and variance factor — \$9.2

Treat each non-overlapping 24-month calendar block as one independent study of the same underlying edge, with effect size its own per-trade Sharpe and variance from \$2.1:

```
for each block w with n_w ≥ 10 trades:
    θ_w = SR_w,    v_w = V_w / n_w,    weight_w = 1 / v_w

θ̄ = Σ weight_w · θ_w / Σ weight_w          # inverse-variance pooled Sharpe
Q  = Σ weight_w · (θ_w - θ̄)²                # ~ χ²(k-1) under a constant true Sharpe
I² = max(0, (Q - (k - 1)) / Q)              # share of dispersion sampling noise cannot explain
```

Q tests whether the block-to-block variation exceeds what sampling noise alone would produce. I^2 says how much of the variation is real. Requiring both a significant Q and $I^2 > 0.5$ means: the heterogeneity is detectable *and* it dominates the noise.

This replaces the `dead_block veto`, which is mostly measuring noise. `dead_block` fires when any non-overlapping block has a negative resampled median, and it fires on **29 of 36 (80.6%)**. Here is why that number is uninformative. Under the null that the true per-trade Sharpe is constant at the population median 0.117, a block of ~200 trades has $SE \approx 1/\sqrt{200} = 0.071$, so

```
P(one block's estimate < 0) = Φ(-0.117/0.071) = Φ(-1.65) ≈ 5%
P(at least one of 7 blocks < 0) = 1 - 0.95⁷ ≈ 30%
```

and with the *out-of-sample* Sharpe rather than the pooled one the per-block probability is far higher still. A genuinely stationary strategy is therefore expected to show a negative block routinely. On the reference population the median number of negative blocks is 1 of 7 — exactly what chance predicts — while the heterogeneity test says the dispersion is beyond noise for only 5 strategies. **`dead_block` converts an ordinary sampling fluctuation into a veto**, and it is the single largest source of false kills in the current system.

On the reference population I^2 spans 0.00–0.71 with median 0.20, and Cochran's p spans 0.002–0.958 with median 0.275. Both have real variance, so both are usable (P7).

Read together with the volatility terciles, not instead of them. Q says *that* the edge moved; `D.regime.buckets` says *what* it moved with. A strategy with high I^2 whose weakness sits entirely in the high-volatility tercile has a describable, possibly fixable regime dependence. One with high I^2 and no tercile pattern has an unexplained one. Both fire V5; the first is worth a note in the kill record, because it is the kind of strategy worth revisiting with a volatility filter.

V6 — The out-of-sample result rests on a handful of trades

Fires when: `best_OOS_trade / gross_profit_OOS > 0.15` or `effective_winners_OOS < 25`

Reference population: kills 0 of 36 (0.0%)

Principle: P4

```
gross_OOS      =  $\sum \max(\text{pnl}_i, 0)$  over OOS trades
best_share     =  $\max(\text{pnl}_i) / \text{gross\_OOS}$ 
effective_winners =  $(\sum_{\{\text{pnl}>0\}} \text{pnl}_i^2) / \sum_{\{\text{pnl}>0\}} \text{pnl}_i$  # 1/Herfindahl of the winners
```

The denominator is **gross profit, never net** (P4). `effective_winners` is the reciprocal Herfindahl index of the winning trades: the number of equally-sized winners that would produce the same concentration. It is the more robust of the two, because it does not depend on a single extreme observation.

On the reference population `best_share` spans 0.021–0.046 and `effective_winners` spans 71–138. The rule fires on nothing. **It is reported with its zero firing rate rather than quietly dropped** (P7): concentration is a real failure mode for strategies with rarer, larger trades, and this population simply does not have it — 66–129 trades per year with a 48% win rate produces broad-based profit. An agent that sees this rule fire should treat it as a genuine anomaly relative to the reference population and say so.

The module's own `outlier` check (`best_trade / net`, threshold 0.25) also fires on nothing here, for a different reason: it happens to use the pooled net, which is large. On the OOS scope alone the same formula fires on 50% of the population from denominator collapse — the P4 trap, live.

V7 — The out-of-sample edge does not survive a symmetric trim

Fires when: `SR_OOS_trimmed ≤ 0`, where the trim removes the top and bottom 1% of OOS trades

Reference population: kills 11 of 36 (30.6%)

Principle: P4

```
sorted = sort(pnl_OOS)
k      = max(1, round(0.01 · n_OOS))
trimmed = sorted[k : n_OOS - k] # k removed from EACH tail
SR_trim = mean(trimmed) / std(trimmed, ddof=1)
```

The trim must be symmetric. Removing the best trades while keeping the worst mechanically drives any Sharpe negative and measures nothing — that was the rejected form in P4's table, which killed 67% of the population with values down to −94.7. Removing the same count from both tails asks the question that was meant: *is the edge broad-based, or does it live in a few extreme observations?*

On the reference population the trim removes 6–12 trades of 292–624. After it, 25 of 36 strategies keep a positive OOS edge and 5 change sign; the median annualised OOS Sharpe falls from 0.199 to 0.129. That shrinkage is the honest measure of how much of this population's out-of-sample performance sits in its tails, and 11 strategies do not survive it at all.

The absolute form (`SR_trim ≤ 0`) is used rather than a retention ratio for the P4 reason: the untrimmed OOS Sharpe is near zero for much of the population, so any ratio to it is uninterpretable.

5. Layer 2 — the score for what survived

Every strategy that fires no veto gets a score in 0–100. The score **rank**s; it does not certify (P1). Its terms are all scale-free, all anchored out-of-sample, and all verified to have real variance on the reference population (P7).

SCORE = 0.35·S1 + 0.25·S2 + 0.15·S3 + 0.15·S4 + 0.10·S5

term	weight	what it measures	formula
S1 persistence	0.35	how good the out-of-sample result actually is	curve(ann_SR_OOS, 0.0, 1.0)
S2 retention	0.25	how much of the in-sample edge survived, and how resolvable the gap is	0.6·curve(SR_OOS/SR_IS, 0.0, 0.8) + 0.4·curve(-z_decay, -2.326, 0.0)
S3 composition	0.15	whether the out-of-sample edge survives resampling and trimming	0.5·curve(pf_5_OOS, 0.90, 1.20) + 0.5·curve(ann_SR_OOS_trimmed, 0.0, 0.8)
S4 cost cushion	0.15	how much worse execution can get	0.5·curve(cost_cushion, 1.5, 4.0) + 0.5·curve(spread_cushion, 2.5, 6.0)
S5 uniformity	0.10	whether the edge is the same across periods, and free of order luck	0.5·curve(1 - I², 0.3, 1.0) + 0.5·curve(-rank, -0.5, -0.05)

pf_5_OOS is the 5th percentile profit factor of a **block** bootstrap over the OOS trades only, block length round(n_OOS^(1/3)) (\$9.3). The Politis–Romano block rule is used because OOS trades are correlated inside a regime and an i.i.d. bootstrap reads more comfortably than the strategy can actually suffer.

Measured spread of each term on the reference population — the P7 check that none is degenerate:

term	min	p5	median	p95	max	std
S1	0.0	0.0	19.9	59.4	76.1	22.1
S2	0.0	0.0	13.3	69.7	88.3	23.7
S3	0.0	0.0	8.1	53.1	69.6	19.8
S4	2.8	10.9	45.0	84.7	91.4	23.2
S5	0.6	13.8	56.0	95.4	100.0	25.3
SCORE	4.3	5.3	25.4	58.9	75.5	19.0

Every term spans most of its range with a standard deviation above 19 points. Compare the current system's sub-scores D (std 0.0) and E (std 0.0).

5.1 Why pf_5_OOS > 1.0 is a score term and not a veto

It is tempting to demand that the 5th percentile profit factor of the OOS trades exceed 1.0 — in one of twenty resamples the strategy must still be profitable. Measured, that rule kills **34 of 36 (94.4%)**, and the reason is arithmetic, not fragility. Net profit over n resampled trades is positive at the 5th percentile only when

SR > 1.645 / √n

which at n_OOS = 440 demands a per-trade Sharpe above 0.078 — reached by 2 of 36 strategies. So pf_5_OOS > 1.0 is the same 95% significance demand as t_OOS ≥ 1.645 in different clothing, and it fails for the same power reason (V1). As a *score* term on a curve from 0.90 to 1.20 it discriminates usefully; as a veto it is a wall.

This is the general lesson for the agent: before using any percentile-based floor as a veto, check what Sharpe it implicitly demands at the sample size in hand. A 5th-percentile floor is a significance test wearing a robustness costume.

5.2 The cut, and how to choose it

After the vetoes, 16 of 36 strategies remain. Their scores and the resulting throughput:

SCORE cut	advance	of population
≥ 25	14	39%
≥ 30	11	31%
≥ 35	10	28%
≥ 40	9	25%
≥ 45	7	19%
≥ 50	7	19%
≥ 55	2	6%

The recommended default is 45, giving 19% advance on the raw databank. The reasoning is not statistical — it cannot be, because the score is a ranking and not a test. It is that the next stage costs real work per strategy, so the cut should be set by what that stage can absorb. **If throughput is the binding constraint, replace the absolute cut with a quota: take the top *k* by SCORE among the non-vetoed.** A quota is more honest than a fixed number when the generation regime changes, because the score's absolute level drifts with the population while its ordering does not.

Below the cut the verdict is **HOLD**, not **KILL**: nothing about these strategies is demonstrably fragile, they are simply not the best of the batch, and a later batch may be worse.

5.3 What the score correlates with, and why that matters

the score against	Spearman ρ
annualised OOS Sharpe	+0.971
the decay z-statistic	-0.660
the current system's composite	+0.612

Compare the current composite against the same quantities (§8.3): +0.690 with annualised OOS Sharpe and **+0.050 ($p = 0.77$) with the decay z-statistic.** The new score is aligned with out-of-sample persistence and with the absence of decay by construction; the old one is aligned with in-sample quality and is blind to decay.

6. Layer 3 — drawdown as a sizing output

By P5, drawdown never vetoes. It produces a number instead.

```
risk_scale      = dd_budget / dd_pct_99
risk_per_trade = risk_scale × 1000      # 1000 USD is config global.risk_per_trade
```

`dd_pct_99` is `A.dd_pct_99`, the 99th percentile of the reordered drawdown distribution — the appropriate input because the question is what the account must survive, not what it happened to survive.

This is a first-order estimate, and it is conservative. Scaling `risk_per_trade` by λ scales every P&L by λ , so dollar drawdown scales exactly by λ , but `dd_pct = max(drop/peak)` divides by a peak that contains the unscaled starting equity. To first order $\text{dd_pct} \approx \lambda \cdot \text{dd_}\$/\text{equity}_0$, linear in λ . The second-order term is favourable: a larger account cushion accumulates as profits do, so the true `dd_pct` grows slightly more slowly than linearly. **For an exact answer, re-run `metrics.paths()` on $\lambda \cdot \text{pnl}$ and read `dd_pct_99` again** — it costs one simulation pass and removes the approximation.

On the reference population, for a 10% budget at the 99th percentile, the implied risk per trade has median \$795 and spans \$370–\$1 908. Read: most of these strategies need to be sized *down* from the \$1 000 the study assumed to fit a 10% budget, and a few can be sized up.

Report alongside it the two numbers that are invariant under resizing and therefore describe the strategy rather than the sizing choice: **Ret/DD** (exactly invariant) and **annualised OOS Sharpe**.

7. The degradation module, in full

The owner's stated priority is avoiding IS→OOS degradation, so this section collects everything bearing on it in one place. Nothing here is new; it is \$1–\$5 re-indexed by that question.

7.1 The five numbers that answer it

Computed on the two scopes from `stream.samples()` , using §2.1 and §2.2:

number	formula	what it tells you	reference population
ann_SR_OOS	§2.2 on the OOS scope	where it landed — the only near-unbiased level estimate	−0.519 ... 0.761, median 0.199
retention	SR_{OOS} / SR_{IS}	how much survived; readable, not decisive	−0.508 ... 0.750, median 0.136
z_decay	$\$4/\sqrt{2}$	whether the gap is bigger than noise	0.46 ... 2.99, median 1.99
PSR_OOS(b)	$\Phi((SR_{OOS} - b) \cdot \sqrt{(n_{OOS}-1)/V_{OOS}})$	probability the true OOS Sharpe beats <code>b</code>	at b=0: 0.127 ... 0.959, median 0.672
ann_SR_OOS_trimmed	V7	whether the landing survives a symmetric trim	−0.690 ... 0.867, median 0.129

The decision content is in `ann_SR_OOS` (level) and `z_decay` (excess decay). Retention is for reading. The trimmed version is the durability check. `PSR_OOS` is the same information as `z_decay` 's numerator expressed as a probability, and is the number to quote to a human.

7.2 Relocating the PSR is the single highest-value change in this document

`family_e` computes the PSR on the **pooled** stream against a benchmark of 0. On the reference population that gives 0.972–1.000 — sub-score E is exactly 100.0 on all 36 strategies. The reason is arithmetic: at $n \approx 1300$ and a pooled per-trade Sharpe of 0.12, $z \approx 0.12 \cdot \sqrt{1299} \approx 4.3$, so $\Phi(z)$ is indistinguishable from 1. **The test is being asked whether an edge that was selected for being positive in-sample is positive in-sample.** It is, always, and 10% of the composite's weight is spent confirming it.

Move the same formula to the OOS scope and it becomes the sharpest single discriminator available:

scope	PSR at benchmark 0, reference population
pooled (as configured today)	0.972 – 1.000, median 0.9996 — zero variance for decisions
OOS only	0.127 – 0.959, median 0.672 — full range

The exemplar `Strategy 17.18.29` illustrates it: pooled PSR 0.99999987, OOS PSR 0.887. The same statistic, the same code, one scope change, and a number that saturated becomes a number that separates.

Do not then gate on `PSR_OOS ≥ 0.90` . That is the 94%-kill wall of \$5.1 again. Use it as the human-readable form of S1 and S2, and keep V1's sign test as the veto.

7.3 How to read the overlay figures without being misled

`degrade.overlay()` draws, per family, the IS and OOS distributions of that family's headline statistic on one axis. As built today they are **not** like-for-like (P3), and the direction of the error differs by family:

family	statistic	how the unequal lengths bias it
A	<code>dd_pct</code>	OOS is flattered : fewer trades, less opportunity to draw down
B, D	<code>net</code>	OOS is penalised : net profit scales with trade count, and OOS has ~43% as many
C (×4)	<code>net</code>	same as B

So the exemplar's Family A overlay — OOS p95 drawdown 6.33% against IS 3.18%, twice as bad — is **worse than it appears**, because the shorter OOS scope should have drawn down *less*. And every Family B and C overlay showing OOS net below IS net is telling you mostly that the OOS window is shorter.

Until §9.4 is implemented: **use the overlays to see shape and direction, and take every quantitative IS/OOS claim from §7.1 instead.** After §9.4, both sides describe `min(n_IS, n_OOS)` trades and the overlays can be read directly.

7.4 A structural caveat the agent must carry

On the reference population the split is a clean chronological walk-forward: in-sample is 2008-01 → 2017-12, out-of-sample is 2018-01 → 2022-12. Two consequences:

- 1. **The OOS scope is one regime draw, not 440 independent ones.** It contains the 2020 liquidity shock and the 2022 rate shock on gold. The trades inside it are correlated through the regime, so an i.i.d. reading of `n_OOS` overstates the evidence. This is why the block bootstrap is specified for `pf_5_OOS` (§5), and it is a second reason not to demand 95% significance of the OOS scope.
- 2. **Degradation and regime change are confounded here and cannot be fully separated.** V5's heterogeneity test and the volatility terciles are what partially separate them: a strategy whose weakness is confined to one volatility tercile has a regime dependence, while one that decayed uniformly has an overfitting signature. Report which of the two the evidence looks like, and never claim more than the data separates.

8. Diagnosis of the current verdict system

This section exists so that the change is auditable rather than asserted. Every number is from the reference population's own `verdict.csv` and `flags.csv`.

8.1 The verdict tiers are unreachable by construction

Sub-score D is **0.0 on all 36 strategies**; sub-score E is **100.0 on all 36**. With `weights = {A:0.25, B:0.25, C:0.20, D:0.20, E:0.10}` the attainable composite is therefore

$$\begin{aligned} \text{composite} &= 0.25 \cdot A + 0.25 \cdot B + 0.20 \cdot C + 0.20 \cdot 0 + 0.10 \cdot 100 \\ &= 0.25 \cdot A + 0.25 \cdot B + 0.20 \cdot C + 10 \end{aligned}$$

The observed maxima are $A = 70.3$, $B = 83.6$, $C = 63.3$, so the highest composite this population can produce is $0.25 \cdot 70.3 + 0.25 \cdot 83.6 + 0.20 \cdot 63.3 + 10 = 61.1$ — **exactly the observed maximum**. The tiers are 80 / 65 / 50, so `STRONG` and `ACCEPTABLE` cannot be reached by any strategy in this databank, whatever it does. The observed verdicts follow mechanically: 33 FAIL, 3 MARGINAL, 0 anything else.

8.2 Why D is always 0 and E is always 100

D is `min` of three terms, the first of which is `curve(windows_ok, 0.70, 1.0)` where `windows_ok` is the fraction of rolling 24-month windows whose 5th-percentile net profit is positive. That fraction spans 9.6%–48.1% on this population, never reaching the 0.70 that scores anything above zero. The reason is \$5.1's arithmetic: `net_5 > 0` in a window of ~215 trades requires a per-trade Sharpe above $1.645/\sqrt{215} = 0.112$, which is *above the population's median pooled Sharpe of 0.117* — so it demands that most windows individually beat the whole sample's central estimate. **The threshold asks for something the population cannot supply**, and because D takes a `min`, one impossible term zeroes the family.

E is `curve(psr, 0.90, 0.95)` and the PSR saturates at 1.0 for the reason in §7.2.

8.3 The composite measures in-sample quality, not robustness

Spearman rank correlations of the current composite against quantities computed independently for this document:

the composite against	ρ	p
cost cushion	+0.858	0.000
annualised IS Sharpe	+0.825	0.000
in-sample per-trade Sharpe	+0.810	0.000
in-sample profit factor	+0.781	0.000
annualised OOS Sharpe	+0.690	0.000
retention	+0.582	0.000
the decay z-statistic	+0.050	0.772
permutation rank of the drawdown	−0.385	0.021

Two readings, both uncomfortable. First, the composite's strongest associations are with in-sample quantities, and its association with the IS→OOS decay — the owner's stated priority — is indistinguishable from zero. Second, the correlation with the permutation rank is **negative**: strategies whose real ordering was luckier score *higher*. The `inflation` term intended to penalise order luck is dominated by the `dd_pct_99` and `ret_dd` terms in the same sub-score, which a lucky ordering improves. The net effect is that the system mildly rewards the thing it was built to punish.

8.4 Which current checks fire, and the verdict on each

check	fires	disposition
inflation_watch	36/36	remove. $\text{inflation} = 1.578 \times \text{luck_ratio}$; the threshold is below the population floor (P6)
inflation (veto)	0/36	remove. Requires $\text{luck_ratio} > 1.90$, above the population ceiling
windows	36/36	remove. Demands a per-window Sharpe above the sample's own median (\$8.2)
dead_block	29/36	replace with V5. ~30% of stationary strategies show a negative block by chance (\$4/V5)
dd_99	21/36	remove. Contradicts P5; becomes the \$6 sizing output
oos_red	30/36, warning only	promote and reformulate. The priority signal, currently unable to veto anything → V1, V2
oos_amber	4/36, warning only	folded into S2
pf_5 (B, pooled)	13/36	re-scope. Pooled is ~70% in-sample; use the OOS scope → S3
psr	0/36	re-scope to OOS (\$7.2), and keep as a score term, not a gate
outlier	0/36	keep as V6 , with the gross-profit denominator (P4)
Family C keep / pf_5	3–4/36 each	keep, reformulated as cushions → V4
high_vol	1/36	keep as supporting evidence for V5; too rare to gate alone
concentration	0/36	keep as a reported diagnostic
sample	0/36	keep → Layer 0

8.5 What the two systems disagree about, concretely

The strategy the current system ranks **first** (Strategy 17.18.29 , composite 61.1, the only one above 55) has retention 0.35, z_decay 1.88 and I^2 0.40. The strategy the new criteria rank first (Strategy 21.30.24 , SCORE 75.5) has annualised OOS Sharpe 0.76, retention 0.68, z_decay 0.66 and I^2 0.00 — the best out-of-sample profile in the databank on every axis — and the current system rates it **FAIL** at composite 47.8, below 12 other strategies.

Conversely Strategy 17.27.43 passes every current veto with composite 50.9, while having the population's worst permutation rank (0.006 — its real ordering beat 99.4% of reshufflings), its highest luck_ratio (1.616), significant heterogeneity ($p = 0.018$, $I^2 = 0.607$), retention 0.14 and z_decay 2.48. It fires V2, V3 and V5. It is the clearest single case of the two systems pointing in opposite directions.

9. What must be added to the code

Five of the rules need numbers the module does not currently emit. Each is small, none needs a new simulation family, and all are closed-form except §9.3. Until each is added, the rules that depend on it return **HOLD** for that axis (§0).

9.1 Per-scope Sharpe with its variance factor — needed by V1, V2, V7, S1, S2, S3

`B.samples[scope]` today carries `n`, the *median bootstrap* Sharpe, `net` and `pf_5`. It does not carry the point-estimate Sharpe, its skew, its kurtosis or its standard error. Add, per scope, the output of the function that already exists:

```
# in run._family_b, inside the loop over stream.samples()
cut = stream.restrict(source, positions)
parts[name] = {..., **significance.psr(cut["pnl"], cfg["family_e"]["psr_benchmark"]),
               "years": float((cut["open"].max() - cut["open"].min())
                             / np.timedelta64(1, "D")) / 365.25}
```

`significance.psr()` returns `sharpe`, `skew`, `kurtosis`, `n` and `psr` — every ingredient of §2.1 and §2.2. The `years` field is what §2.2 needs to annualise on the scope's own trade rate.

9.2 Per-block Sharpe and variance factor — needed by V5

`familyd._slices()` stores `median_net`, `net_5`, `pf_5` and `n` per window. Add the point estimates — no simulation, so the cost is negligible:

```
rows.append(..., **{f"sr_{k}": v for k, v in
                    significance.psr(source["pnl"][pos], 0.0).items()})
```

Then compute Cochran's Q and I^2 over the **non-overlapping** blocks only (the rolling windows overlap, so their estimates are not independent and Q is invalid on them). The formula is in V5.

9.3 OOS-scope block bootstrap — needed by S3

`B.samples[scope]["pf_5"]` uses `iid_bootstrap`. Add a `block_bootstrap` run on the same scope at `config.stationary_block(n_scope)`, and read `pf_5` from it. The justification is §7.4: OOS trades are correlated through the regime, and the i.i.d. bootstrap reads more comfortably than the strategy can suffer. Keep the i.i.d. number too — the gap between them is itself informative.

9.4 Equal-length IS/OOS resampling — needed to make the overlays readable (§7.3)

`degrade.scope_shape()` resamples each scope at its own length. Give it a target length and use `min(n_IS, n_OOS)` for both:

```
def scope_shape(source, positions, kind, model, block, metric, cfg, size=None):
    ...
    got = engine.sequential(engine.payload(cut), kind, model, block,
                                cfg["global"]["n_sims"], cfg, size=size)
```

This requires `draws.*` to accept a `size` that differs from the number of trades available, which `iid_bootstrap` and `block_bootstrap` already support naturally (they draw with replacement) and `iid_shuffle` and `block_shuffle` cannot (they are permutations). **So the equal-length comparison is available for the bootstrap families B, C and D, and not for A.** For Family A, compare instead on the length-normalised statistic `dd_pct /  $\sqrt{n}$` , which is the §P3 scaling, and say that is what is plotted.

9.5 Cost and spread totals — convenience for V4

`V4` can be recovered in closed form from Family C's `keep` (§4/V4), but storing the two sums removes the algebra and the modelling asymmetry from the reading path:

```
# in run.analyse
"costs": {"charged": float(source["cost"].sum()),
         "spread": float(source["spread"].sum()),
         "cushion": float(source["pnl"].sum() / source["cost"].sum()),
         "spread_cushion": float(source["pnl"].sum() / source["spread"].sum())}
```

9.6 One recalibration run

After §9.1–§9.5, re-run the whole reference databank and re-measure every firing rate in this document. Four of the constants here were set by measuring candidate rules against this population and rejecting the versions that killed 50–94% of it (P4, §5.1); the same measurement must be repeated whenever the numbers feeding the rules change. The companion document's Appendix B notes that the Family D columns of the last full run predate three configuration

changes and were never re-measured — that is exactly the gap this step closes.

10. The decision procedure

The whole document as one algorithm. The agent executes this and nothing else.

```
INPUT: result (the full result object for one strategy), cfg, dd_budget

# ---- Layer 0: is the analysis fit to decide on? -----
for check in [A1..A8]:
    if failed(check):
        record(check)
        if check == A6: return HOLD, "A6", "draw model is broken - do not decide"
        if check in [A1, A2, A3]: return HOLD, check, value
        # A4, A5, A7, A8 disable specific axes rather than the whole decision
        disable_axes(check)

# ---- compute the primitives (§2) -----
for scope in [IS, OOS]:
    SR, g3, g4, V, SE, n, years <- §2.1, §2.2
    ann_SR[scope] = SR[scope] * sqrt(n[scope] / years[scope])

z_decay = (SR[IS] - SR[OOS]) / sqrt(V[IS]/n[IS] + V[OOS]/n[OOS])
retention = SR[OOS] / SR[IS]
SR_trim = symmetric 1% trim of OOS trades (§4/V7)
rank_iid = result.A.runs["iid_shuffle"].dd_pct.rank (§4/V3)
rank_blk = result.A.runs["block_shuffle/<max>"].dd_pct.rank
Q, I2 = Cochran over result.D.nonoverlapping (§4/V5)
cushions = from result.costs, or in closed form from Family C (§4/V4)

# ---- Layer 1: the vetoes (§4) -----
fired = []
if SR[OOS] <= 0 or PF[OOS] <= 1.0: fired += [V1]
if z_decay > 2.326 and ann_SR[OOS] < 0.50: fired += [V2]
if rank_iid <= 0.05 and rank_blk <= 0.05: fired += [V3]
if cost_cushion < 1.5 or spread_cushion < 2.0: fired += [V4]
if Q_pvalue < 0.05 and I2 > 0.50: fired += [V5]
if best_share_gross > 0.15 or eff_winners < 25: fired += [V6]
if SR_trim <= 0: fired += [V7]

if any axis needed by a veto was disabled at Layer 0 and that veto did not fire:
    that veto's outcome is HOLD, not PASS # absence of evidence is not evidence

# ---- Layer 2: the score (§5) -----
SCORE = 0.35*S1 + 0.25*S2 + 0.15*S3 + 0.15*S4 + 0.10*S5

# ---- Layer 3: sizing (§6) -----
risk_scale = dd_budget / result.A.dd_pct_99
risk_per_trade = risk_scale * cfg.global.risk_per_trade

# ---- the verdict -----
if fired non-empty: return KILL, binding = fired[0], evidence = fired
elif any HOLD outcome: return HOLD, binding = that check
elif SCORE >= 45: return ADVANCE, evidence = SCORE and its five terms
else: return HOLD, "below the batch cut, not fragile"
```

Reporting. For every strategy, output the decision, the binding rule with its value and limit, every rule that fired, the five sub-scores, `risk_per_trade`, and the five degradation numbers of §7.1. For a KILL, name **all** vetoes that fired, not the first — a strategy killed by V2 alone is a candidate for revisiting with a regime filter, while one killed by V2, V3 and V5 together is not.

10.1 What this produces on the reference population

decision	count	share
ADVANCE (no veto, SCORE ≥ 45)	7	19%
HOLD (no veto, SCORE < 45)	9	25%
KILL (≥ 1 veto)	20	56%

against the current system's 33 FAIL, 3 MARGINAL, 0 pass. The seven that advance, ranked:

rank	strategy	SCORE	ann OOS SR	retention	z decay	I ²	cost cushion	current verdict
1	Strategy 21.30.24	75.5	0.76	0.68	0.66	0.00	2.50	FAIL (47.8)
2	Strategy 21.28.44	72.8	0.76	0.75	0.46	0.03	2.51	FAIL (48.2)
3	Strategy 20.5.40	54.3	0.54	0.53	0.90	0.00	2.21	FAIL (43.4)
4	Strategy 1.17.44	52.5	0.52	0.54	0.84	0.06	2.11	FAIL (38.8)
5	Strategy 24.17.35	52.2	0.44	0.41	1.15	0.00	2.68	FAIL (42.9)
6	Strategy 17.18.29	51.9	0.53	0.35	1.88	0.40	3.57	MARGINAL (61.1)
7	Strategy 21.35.36	50.4	0.49	0.47	1.03	0.10	2.24	FAIL (44.3)

Read this honestly. Seven strategies advance, and the best of them has an annualised out-of-sample Sharpe of 0.76 over a single five-year regime — respectable, not remarkable, and not statistically certified by that sample ($t_{OOS} = 1.73$). The criteria are not claiming these seven are good. They are claiming these seven are the ones whose *out-of-sample* performance is positive, broad-based, not order-luck, not regime-confined, and able to absorb worse costs — and that the rest either fail one of those outright or rank below them. That is the strongest claim this data supports (P1).

11. What Monte Carlo may assume from upstream

The module's README places this study **after** the IS/OOS decay test and the cross-market retest. That changes what the criteria are for, and it changes every firing rate quoted above — so it is stated here explicitly rather than left implicit.

11.1 The reference population was not filtered, and the rates above reflect that

All firing rates in §3–§5 are measured on the **raw** 36-strategy databank. That databank was clearly not pre-filtered: 30 of 36 arrive with retention below 0.40, and 10 arrive with a negative out-of-sample Sharpe. **Every rate above is therefore an upper bound on what the rules will do to a properly filtered input.**

11.2 What survives the decay test, reconstructed

The decay test, as the companion document describes it, keeps a strategy when its OOS Sharpe is distinguishable from zero on its own standard error, it won in most OOS years separately, and no single quarter carries the result. Reconstructing those three legs on the same 36 strategies:

leg of the decay test	passes
OOS t-statistic ≥ 1.0	7 / 36
majority of the 5 OOS years positive	19 / 36
no single quarter above 50% of OOS profit, measured against gross	36 / 36
all three	7 / 36 (19%)

The t-statistic leg is the only one that binds, and where it is set decides everything downstream:

upstream cut	reach Monte Carlo	vetoed fire on	ADVANCE at SCORE ≥ 45
$t_{\text{OOS}} \geq 0$ (sign only)	18 / 36	6 of 18 — V2×4, V5×3, V3×2, V7×1	7 of 18 (39%)
$t_{\text{OOS}} \geq 0.5$	13 / 36	3 of 13 — V2×3, V5×2	7 of 13 (54%)
$t_{\text{OOS}} \geq 1.0$	7 / 36	0 of 7	6 of 7 (86%)
$t_{\text{OOS}} \geq 1.645$	2 / 36	0 of 2	2 of 2 (100%)

At $t_{\text{OOS}} \geq 1.0$ the seven vetoes fire on nothing. The decay test has already removed every strategy they would have caught, and Monte Carlo's contribution collapses to the ranking score.

11.3 The recommended division of labour

The two stages should not both be level filters. They see different things:

- **The decay test sees the level and the calendar** — is there an out-of-sample edge at all, was it present across years. It cannot see order luck, cost cushion, regime heterogeneity or tail dependence, because those need the resampling this module does.
- **Monte Carlo sees the structure** — V3 (order luck), V4 (cost cushion), V5 (regime heterogeneity), V6 (concentration) and V7 (tail dependence) are questions no upstream stage answers, and they are why a strategy should still pass through here after being kept.

Recommendation: cut upstream at $t_{\text{OOS}} \geq 0.5$, not at 1.0. Two reasons, both measured:

1. A cut at 1.0 discards 29 of 36 on a single statistic computed on one regime draw. By §4/V1's power arithmetic, $P(t_{\text{OOS}} < 1.0)$ for a strategy whose *true* annualised OOS Sharpe is 0.75 is about 28% — so roughly one good strategy in four is thrown away by that leg alone, with no chance to be seen by any of the structural tests.
2. At 0.5 the vetoes still do real work (3 of 13), so the two stages are complementary rather than nested. At 1.0 they are nested, and the second stage is decorative.

If instead the upstream cut stays at 1.0, then **V1 and V2 should be dropped from this document** — they are dead weight on that input — and the criteria become V3–V7 plus the score. Keeping a veto that provably cannot fire violates P7 as surely as keeping one that always fires.

11.4 A defect in the upstream test worth fixing

The quarter-concentration leg is the P4 trap (§1) in another module. Measured both ways on the same 36 strategies:

best OOS quarter, as a share of...	min	median	max	fails a 50% rule
net OOS profit	0.22	0.71	42.64	17 / 36
gross OOS profit	0.15	0.23	0.38	0 / 36

Over 20 quarters an even split gives 5% each, so a best quarter at 23% of gross is mild concentration and 0 of 36 strategies are anywhere near a 50% rule. With the **net** denominator the same quantity reaches 42.64 — a strategy whose best quarter is forty-two times its total out-of-sample profit, which says only that its total is near zero. **A 50% rule on the net denominator rejects 17 of 36 strategies for having small net profit while calling it concentration.**

If the decay test uses the net denominator, it is rejecting on an artefact and the fix is one division. This is a finding about that module, not this one, and it is recorded here because it was measured here.

Appendix A — the reference population, strategy by strategy

All 36 strategies of databank `Results`, project `XAUUSD`, export `2026-09-03`. Sorted by decision then SCORE. `fired` lists the vetoes; `old tier` / `old comp` are the current system's verdict for comparison. Retention, `z decay`, `perm rank`, `luck`, `I²`, `het p` and the cushions were computed directly from the trade files for this document; `SCORE` and `decision` are \$5 and \$10.

strategy	n IS	n OOS	SR IS	SR OOS	ann OOS	trim ann OOS	retention	z decay	perm rank	luck	l ²	het p	cost cush	spread cush	OOS pf5	SCORE	decision	fired	old tier	old comp
Strategy 21.30.24	929	435	0.1191	0.0813	0.76	0.79	0.68	0.66	0.146	1.27	0.00	0.551	2.50	5.56	1.022	75.5	ADVANCE	—	FAIL	47.8
Strategy 21.28.44	780	371	0.1169	0.0877	0.76	0.87	0.75	0.46	0.532	0.98	0.03	0.403	2.51	5.26	1.018	72.8	ADVANCE	—	FAIL	48.2
Strategy 20.5.40	1,073	496	0.1018	0.0539	0.54	0.50	0.53	0.90	0.462	1.02	0.00	0.786	2.21	5.30	0.947	54.3	ADVANCE	—	FAIL	43.4
Strategy 1.17.44	1,012	458	0.1001	0.0543	0.52	0.43	0.54	0.84	0.221	1.20	0.06	0.379	2.11	4.11	0.928	52.5	ADVANCE	—	FAIL	38.8
Strategy 24.17.35	958	489	0.1083	0.0439	0.44	0.58	0.41	1.15	0.227	1.19	0.00	0.652	2.68	5.66	0.918	52.2	ADVANCE	—	FAIL	42.9
Strategy 17.18.29	690	292	0.1997	0.0697	0.53	0.69	0.35	1.88	0.346	1.10	0.40	0.124	3.57	6.12	0.926	51.9	ADVANCE	—	MARGINAL	61.1
Strategy 21.35.36	862	411	0.1151	0.0535	0.49	0.56	0.47	1.03	0.329	1.12	0.10	0.353	2.24	4.46	0.949	50.4	ADVANCE	—	FAIL	44.3
Strategy 24.19.23	1,060	527	0.1499	0.0449	0.46	0.61	0.30	1.97	0.537	0.98	0.47	0.078	3.30	6.01	0.934	44.4	ADVANCE	—	FAIL	50.6
Strategy 24.22.43	1,025	520	0.1087	0.0320	0.33	0.47	0.29	1.41	0.099	1.33	0.01	0.418	2.51	5.00	0.885	43.1	ADVANCE	—	FAIL	38.6
Strategy 22.2.30	842	452	0.1090	0.0280	0.27	0.24	0.26	1.39	0.372	1.08	0.00	0.958	2.84	4.62	0.871	35.4	HOLD-LOW	—	FAIL	37.1
Strategy 23.32.31	974	504	0.0989	0.0203	0.20	0.32	0.21	1.42	0.205	1.22	0.00	0.481	2.20	4.40	0.868	32.4	HOLD-LOW	—	FAIL	33.4
Strategy 7.13.36	988	497	0.0981	0.0181	0.18	0.28	0.18	1.46	0.452	1.03	0.00	0.661	1.90	5.38	0.846	29.0	HOLD-LOW	—	FAIL	35.0
Strategy 14.16.34	932	467	0.1075	0.0233	0.23	0.10	0.22	1.50	0.548	0.97	0.14	0.325	1.97	4.96	0.889	27.2	HOLD-LOW	—	FAIL	36.9
Strategy 17.25.27	1,204	566	0.1132	0.0201	0.21	0.16	0.18	1.85	0.095	1.35	0.48	0.072	2.39	4.44	0.876	27.2	HOLD-LOW	—	FAIL	35.3
Strategy 7.14.26	890	453	0.0970	0.0051	0.05	0.01	0.05	1.60	0.055	1.44	0.00	0.638	1.88	4.36	0.815	21.0	HOLD-LOW	—	FAIL	29.0
Strategy 17.29.43	729	337	0.1436	0.0004	0.00	0.06	0.00	2.21	0.170	1.24	0.29	0.210	2.16	4.19	0.783	13.5	HOLD-LOW	—	FAIL	40.4
Strategy 17.17.41	901	439	0.1757	0.0439	0.41	0.34	0.25	2.38	0.071	1.38	0.42	0.114	3.00	4.93	0.910	39.1	KILL	V2	MARGINAL	55.9
Strategy 19.21.30	725	336	0.1500	0.0269	0.22	0.29	0.18	1.89	0.029	1.49	0.00	0.430	2.49	4.99	0.838	34.0	KILL	V3	FAIL	45.1
Strategy 17.27.43	760	370	0.1761	0.0253	0.22	0.13	0.14	2.48	0.006	1.62	0.61	0.018	3.52	5.57	0.845	29.9	KILL	V2,V3,V5	MARGINAL	50.9
Strategy 18.28.28	892	425	0.1930	0.0249	0.23	0.12	0.13	2.99	0.218	1.18	0.63	0.013	3.37	5.23	0.876	26.7	KILL	V2,V5	FAIL	58.8
Strategy 16.20.43	863	424	0.1803	0.0248	0.23	0.25	0.14	2.70	0.144	1.26	0.64	0.011	2.66	4.09	0.864	24.2	KILL	V2,V5	FAIL	53.7
Strategy 21.19.37	1,165	557	0.1322	0.0158	0.17	-0.06	0.12	2.35	0.006	1.66	0.30	0.197	2.42	4.17	0.862	22.2	KILL	V2,V3,V7	FAIL	43.3
Strategy 15.24.36	1,131	543	0.1248	0.0146	0.15	-0.00	0.12	2.19	0.070	1.39	0.28	0.212	2.34	4.07	0.857	21.8	KILL	V7	FAIL	42.9
Strategy 18.27.27	1,312	624	0.1290	0.0173	0.19	0.15	0.13	2.35	0.182	1.24	0.49	0.069	2.33	3.97	0.878	21.4	KILL	V2	FAIL	44.7
Strategy 22.16.41	884	396	0.1453	0.0109	0.10	0.13	0.07	2.23	0.495	1.00	0.71	0.002	3.15	5.91	0.829	18.7	KILL	V5	FAIL	46.4
Strategy 18.22.27	737	349	0.1522	-0.0044	-0.04	-0.06	-0.03	2.47	0.275	1.15	0.04	0.399	2.55	4.92	0.768	15.6	KILL	V1,V2,V7	FAIL	47.8
Strategy 24.26.26	836	431	0.1056	0.0064	0.06	-0.03	0.06	1.69	0.422	1.06	0.00	0.552	1.57	4.01	0.813	15.2	KILL	V7	FAIL	29.3
Strategy 22.13.34	809	384	0.1231	-0.0013	-0.01	0.08	-0.01	2.01	0.702	0.86	0.52	0.053	2.99	5.31	0.805	13.9	KILL	V1	FAIL	37.9
Strategy 6.22.30	823	437	0.0995	-0.0074	-0.07	-0.14	-0.07	1.80	0.434	1.05	0.27	0.225	1.53	4.29	0.770	10.0	KILL	V1,V7	FAIL	25.6
Strategy 10.15.25	1,057	516	0.1025	-0.0091	-0.09	0.00	-0.09	2.10	0.213	1.21	0.21	0.269	1.75	2.82	0.807	9.1	KILL	V1	FAIL	31.1
Strategy 6.13.33	837	430	0.0954	-0.0249	-0.23	-0.36	-0.26	2.03	0.500	1.00	0.17	0.301	1.08	3.32	0.759	6.8	KILL	V1,V4,V7	FAIL	23.1

strategy	n IS	n OOS	SR IS	SR OOS	ann OOS	trim ann OOS	retention	z decay	perm rank	luck	l ²	het p	cost cush	spread cush	OOS pf5	SCORE	decision	fired	old tier	old comp
Strategy 16.21.23	720	330	0.1331	-0.0076	-0.06	-0.02	-0.06	2.14	0.486	1.01	0.52	0.050	1.93	3.51	0.749	5.7	KILL	V1,V5,V7	FAIL	39.3
Strategy 20.23.30	768	370	0.1242	-0.0520	-0.45	-0.40	-0.42	2.82	0.462	1.03	0.20	0.281	1.59	3.17	0.675	5.7	KILL	V1,V2,V7	FAIL	28.0
Strategy 2.13.39	966	490	0.1030	-0.0524	-0.52	-0.69	-0.51	2.81	0.811	0.78	0.26	0.228	1.13	3.57	0.707	5.4	KILL	V1,V2,V4,V7	FAIL	23.6
Strategy 8.11.37	812	419	0.1167	-0.0301	-0.28	-0.41	-0.26	2.46	0.701	0.87	0.44	0.095	1.37	3.99	0.733	5.0	KILL	V1,V2,V4,V7	FAIL	24.3
Strategy 1.25.34	946	457	0.0976	-0.0348	-0.33	-0.46	-0.36	2.36	0.865	0.74	0.15	0.312	1.12	2.70	0.705	4.3	KILL	V1,V2,V4,V7	FAIL	21.3

Appendix B — population quantiles of every gated quantity

quantity	min	p5	p25	median	p75	p95	max
n_00S	292	334	393	438	496	559	624
sr_IS	0.0954	0.0974	0.1029	0.1168	0.1440	0.1834	0.1997
sr_00S	-0.0524	-0.0391	-0.0020	0.0191	0.0350	0.0726	0.0877
ann_IS	0.873	0.913	1.012	1.095	1.338	1.671	1.824
ann_00S	-0.5189	-0.3624	-0.0175	0.1995	0.3489	0.5936	0.7614
oos_ann_sym	-0.6904	-0.4199	-0.0270	0.1291	0.3640	0.7174	0.8667
t_00S	-1.142	-0.806	-0.039	0.445	0.781	1.341	1.733
hl	-0.5084	-0.3724	-0.0149	0.1361	0.2660	0.5774	0.7504
z_degrade	0.464	0.793	1.446	1.991	2.354	2.811	2.986
p_degrade	0.0014	0.0025	0.0093	0.0233	0.0741	0.2146	0.3213
rank	0.006	0.023	0.146	0.302	0.488	0.729	0.865
luck_ratio	0.739	0.843	1.007	1.137	1.265	1.522	1.662
skew_ratio	1.535	1.547	1.568	1.578	1.585	1.604	1.622
het_I2	0.000	0.000	0.005	0.203	0.451	0.632	0.710
het_p	0.002	0.012	0.091	0.275	0.421	0.692	0.958
het_n_neg	0.000	0.000	1.000	1.000	1.000	2.000	3.000
cost_cushion	1.083	1.125	1.895	2.334	2.666	3.405	3.568
spread_cushion	2.695	3.086	4.052	4.448	5.272	5.932	6.119
oos_pf5	0.675	0.707	0.800	0.859	0.894	0.966	1.022
oos_best_gross	0.0213	0.0220	0.0265	0.0278	0.0327	0.0374	0.0459
oos_eff_winners	71	73	92	105	115	135	138
oos_win_rate	0.407	0.433	0.450	0.483	0.491	0.514	0.528
pf_IS	1.307	1.334	1.353	1.403	1.526	1.729	1.804
pf_00S	0.866	0.893	0.994	1.055	1.104	1.225	1.269
dd_pct_99	0.0524	0.0544	0.0908	0.1258	0.1715	0.2291	0.2701
tpy	66	71	82	89	99	116	129

Appendix C — every constant in one place

id	constant	value	fires on the reference population	tune?
A2	minimum <code>n_00S</code> to decide	200	0/36	from <code>confidence.percentile</code> , not free
A2	minimum <code>n_00S</code> to say anything	60	0/36	no
A5	<code>stability.worst_spread</code> tolerance	0.10	—	existing config
A7	volatility coverage floor	0.95	0/36	no
V1	OOS Sharpe floor	0.0	10/36	no — it is a sign test
V1	OOS profit factor floor	1.0	10/36	no
V2	<code>z_decay</code> threshold	2.326 (p<0.01)	11/36 with the floor	the sensitive one: 24/36 at p<0.05, 4/36 at p<0.005
V2	landing floor <code>ann_SR_00S</code>	0.60	inert 0.50–0.80 here (\$4/V2)	yes, but it does nothing on this population
V3	permutation rank threshold	0.05	3/36	no — it is a p-value
V4	<code>cost_cushion</code> floor	1.5	4/36	yes, with the asset
V4	<code>spread_cushion</code> floor	2.0	0/36	yes, with the asset
V5	Cochran <code>Q</code> p-value	0.05	5/36 (with I ²)	no — it is a p-value
V5	<code>I²</code> threshold	0.50	5/36 (with Q)	yes
V6	best trade / gross profit	0.15	0/36	yes, with trade frequency
V6	effective winners floor	25	0/36	yes, with trade frequency
V7	symmetric trim fraction	1% each tail	11/36	no
S1–S5	weights	0.35/0.25/0.15/0.15/0.10	—	yes
\$5.2	SCORE cut	45	7/36 advance	yes — set by throughput
\$6	drawdown budget	caller’s choice	—	yes

Removed from the current system: `dd_99` (P5), `inflation` and `inflation_watch` (P6), `windows` (\$8.2), `dead_block` (superseded by V5), pooled `psr` as a gate (\$7.2).

Appendix D — the traps, as a checklist

Run through this before trusting any number in a Monte Carlo report.

1. **Is the denominator near zero?** Any ratio to OOS net profit is suspect. Use gross profit, or an absolute form. (P4 — four rules were rejected on this.)
2. **Are the two samples the same length?** $n_{IS} \approx 2.3 \cdot n_{OOS}$. Net profit and drawdown are not comparable across them without normalisation. (P3)
3. **Is this percentile floor secretly a significance test?** A 5th-percentile floor on net or profit factor demands $SR > 1.645/\sqrt{n}$. Compute what that is at the sample size in hand before believing it measures robustness. (§5.1)
4. **Is the statistic pooled when it should be scoped?** A pooled number on this population is ~70% in-sample, and in-sample is selection-inflated. (P2)
5. **Does this threshold fire on everything, or nothing?** Check the firing rate before reading any meaning into the flag. (P7 — `inflation_watch` 36/36, `windows` 36/36, `inflation` 0/36)
6. **Is the number annualised on the right trade rate?** Per scope, never pooled. (§2.2)
7. **Is a negative period a finding, or arithmetic?** ~30% of stationary strategies show a negative 24-month block by chance. Test the heterogeneity; do not count the negatives. (V5)
8. **Is a low drawdown skill, or the ordering?** Read the permutation rank, not the inflation ratio. (P6, V3)
9. **Are the two Family C keep values comparable?** No. `cost_shock` charges the increment, `spread_widen` charges the whole spread on top of a net P&L. Compare cushions. (V4)
10. **Is drawdown being used to disqualify?** It must not be. It sets the risk per trade. (P5, §6)